

Towards SVD "Singular Vector Decomposition"

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Oct 2025

What is a Matrix?

Definition

A matrix is a rectangular array (a grid) of numbers, symbols, or expressions, arranged in **rows** and **columns**.

- Matrices are the fundamental objects of Linear Algebra.
- They are used to represent data, systems of equations, and linear transformations.

Visual Representation

We can "visualize" a matrix by thinking of each number as a value in a grid cell. The video uses brightness/color to represent the value.

Matrix Dimensions: $m \times n$

- A matrix with m rows and n columns is called an $m \times n$ matrix (read: "m by n").
- The element at row i and column j is denoted a_{ij} .

- $m = 3$ (rows)
- $n = 4$ (columns)
- $a_{23} = 0$

Example: 3×4 Matrix

$$A = \begin{pmatrix} 5 & 2 & 9 & 1 \\ 8 & 3 & 0 & 6 \\ 4 & 7 & 1 & 3 \end{pmatrix}$$

- ☐ IM: Identity Matrix
- ☐ Scalar Matrix
- ☐ 'IM', but off by one
- ☐ 'IM', but with negative entries
- ☐ Diagonal Matrix
- ☐ Square Zero Matrix
- ☐ Shear Matrix
- ☒ Orthogonal Matrix
- ☐ Projection Matrix
- ☐ Invertible Matrix

Focus 1: Square Matrix

Definition

A **square matrix** has the same number of rows and columns ($m = n$).

- This is a $n \times n$ matrix.
- Square matrices are extremely important and have special properties (like determinants and inverses).
- The elements $a_{11}, a_{22}, \dots, a_{nn}$ form the **main diagonal**.

Example: 3×3 Square Matrix

$$A = \begin{pmatrix} \mathbf{9} & 2 & 7 \\ 1 & \mathbf{5} & 3 \\ 8 & 4 & \mathbf{1} \end{pmatrix}$$

The main diagonal is (9, 5, 1).

Focus 2: Row and Column Vectors

Row Vector

A matrix with only **one row** ($m = 1$). A $1 \times n$ matrix.

$$\mathbf{v} = (1 \quad 2 \quad 3 \quad 4)$$

Column Vector

A matrix with only **one column** ($n = 1$). A $m \times 1$ matrix.

$$\mathbf{u} = \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix}$$

- Vectors are the fundamental building blocks of matrices.
- In many contexts, "vector" implies "column vector".

Focus 3: The Zero Matrix (0)

Definition

A matrix (of any dimension $m \times n$) where **all** elements are equal to zero.

- It is the additive identity: $A + 0 = A$.
- In the video's visualization, this would be a grid of all one color (e.g., all black, representing '0').

Example: 2×3 Zero Matrix

$$0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Focus 4: The Identity Matrix (I)

Definition

A **square matrix** ($n \times n$) with **ones** on the main diagonal and **zeros** everywhere else.

- It is the multiplicative identity: $AI = IA = A$.
- This is one of the most important matrices in linear algebra.

Example: 3×3 Identity Matrix

$$I_3 = \begin{pmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{1} \end{pmatrix}$$

Visualizing the Identity Matrix

- The video visualizes I as a dark grid (representing 0) with a bright, shining line of 1s along the main diagonal.
- This "line of ones" from top-left to bottom-right is the signature visual.

Key Property

Multiplying a vector by an identity matrix does not change the vector. It represents a transformation that does "nothing".

Focus 5: Diagonal Matrix

Definition

A **square matrix** where all elements **off** the main diagonal are zero.

- The elements on the diagonal (a_{ii}) can be any number, including zero.
- This is a generalization of the Identity Matrix.

Example: 3×3 Diagonal Matrix

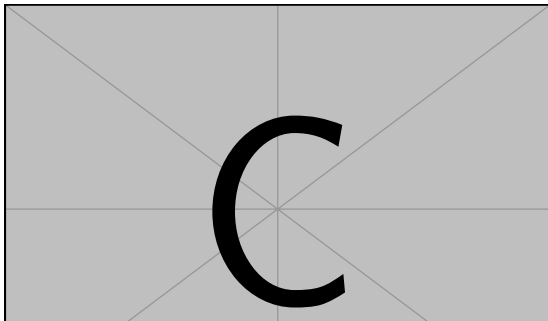
$$D = \begin{pmatrix} 7 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Focus 6: Triangular Matrices

- These are square matrices defined by having all-zero elements on one side of the main diagonal.
- They are extremely important in computation, especially for solving systems of linear equations (e.g., in LU Decomposition).

Two Types

- 1 Upper Triangular Matrix (U)
- 2 Lower Triangular Matrix (L)



Upper Triangular Matrix (U)

Definition

A square matrix where all elements **below** the main diagonal are zero ($a_{ij} = 0$ if $i > j$).

- The non-zero elements form a "triangle" in the upper-right.
- Visually, there is a "sea of zeros" in the bottom-left.

Example

$$U = \begin{pmatrix} 5 & 2 & 8 \\ \mathbf{0} & 1 & 4 \\ \mathbf{0} & \mathbf{0} & 3 \end{pmatrix}$$

Lower Triangular Matrix (L)

Definition

A square matrix where all elements **above** the main diagonal are zero ($a_{ij} = 0$ if $i < j$).

- The non-zero elements form a "triangle" in the lower-left.
- Visually, there is a "sea of zeros" in the upper-right.

Example

$$L = \begin{pmatrix} 5 & \mathbf{0} & \mathbf{0} \\ 7 & 1 & \mathbf{0} \\ 2 & 9 & 3 \end{pmatrix}$$

Focus 7: Symmetric Matrix

Definition

A **square matrix** that is equal to its own transpose ($A = A^T$).

- This means that the element at a_{ij} is equal to the element at a_{ji} for all i and j .
- The matrix is a "mirror image" of itself across the main diagonal.

Importance

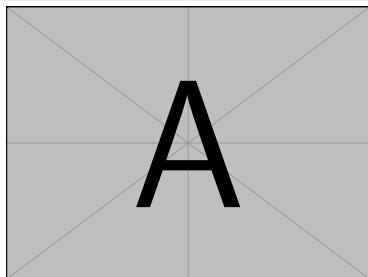
Symmetric matrices are fundamental in physics, data science (covariance matrices), and machine learning (kernel matrices).

Visualizing a Symmetric Matrix

- The main diagonal can be anything.
- Every element in the upper triangle has a matching partner in the lower triangle.

Example: 3×3 Symmetric Matrix

$$S = \begin{pmatrix} 9 & 2 & 7 \\ 2 & 5 & 4 \\ 7 & 4 & 1 \end{pmatrix}$$



Visual: A mirror-image pattern across the diagonal.

Conclusion: Part 1

- This video introduces the "cast of characters" in linear algebra.
- We can classify matrices by their shape (square, vector) or their structure (zero, identity, diagonal, triangular, symmetric).

The Big Idea

Visually identifying these structures is the first step. The next step is to understand what these matrices *do* (i.e., their operations and transformations).

Thank you.

(End of Chapter 1 Summary)

Recap: Linear Transformation

- A matrix is a **linear transformation** of space.
- The transformation is determined by the landing spots of the **standard basis vectors** ($\hat{i}$ and $\hat{j}$).

Standard Matrix Form

- The columns of the matrix are the new coordinates of the transformed basis vectors.

The Starting Point

- All visualizations begin from the **Identity Matrix**:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- This represents **no transformation**.

Introducing Shear Transformation

- Shear matrices produce a **shear transformation**.
- This effect is best described visually.
- The example matrix $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is a shear matrix.

Visualizing 2D Shear

- Vectors **above the origin** move right.
- Vectors **below the origin** move left.

Parallel Movement

- The vectors move **parallel to the x-axis** (in this specific example).

Shape Distortion

- A starting square is **slanted into a parallelogram**.

2D Shear Directions

- In 2D, a shear transformation can be applied in four ways:
 - ▶ Two parallel to the x-axis
 - ▶ Two parallel to the y-axis

Key Property: Area Preservation

- A shear transformation **preserves the area** of the object.

Shear vs. Scalar Matrix

- **Shear:** Preserves **area**, changes **shape**.
- **Scalar (from Part 1):** Preserves **shape**, changes **area**.

3D Shear Concept

- Imagine a **plane slicing through the origin** and a corresponding direction.
- Vectors move **parallel to the plane** while slanting towards that direction.

Orthogonal Matrix Definition

- A **square matrix** where:
 - ① Every column vector is a **unit vector**.
 - ② All column vectors are **orthogonal** to each other.

Dissecting: Column Vector

- A matrix can be thought of as a **collection of vectors** (the columns).

Dissecting: Unit Vector

- A vector is a **unit vector** if its **magnitude is one** (length of the arrow is one).

Dissecting: Orthogonal

- The formal definition of orthogonality: The **dot product** between two vectors is **zero**.

Visualizing Orthogonality

- Visually, a zero dot product means the two vectors are **perpendicular** (at a 90-degree angle).

2D Orthogonal Example

- A sample orthogonal matrix is shown in the video, confirming its column vectors are unit and orthogonal.

The Orthogonal Transformation

- An orthogonal matrix produces a **pristine rotation**.
- There is **no scaling, stretching, shearing, or reflection**.

Significance of Rotation

- Finding the correct rotation direction is a crucial step in understanding the complexity of linear transformation.
- An orthogonal matrix **always produces a rotation**.

3D Rotation

- A 3×3 orthogonal matrix produces a rotation in three dimensions.
- This can occur around any axis (e.g., the z-axis).

Composition of Transformations

- Matrix-to-matrix multiplication is a **composition of sequential transformations**.

Setup: Two Rotations

- **Matrix A:** Rotation around the **x-axis** by 60° .
- **Matrix B:** Rotation around the **z-axis** by 45° .

The Product Matrix $C = B \times A$

- Multiplying the matrices yields C.
- The transformation is A (x-rotation) first, then B (z-rotation).

Encapsulation

- The resulting matrix C encapsulates the two sequential transformations into **just one rotation**.

Order Reversal

- Reversing the sequence: B first (z-rotation), then A (x-rotation).
- The final position of the object is **different**.

Non-Commutativity

- This proves that matrix composition (multiplication) is **not commutative**:

$$B \times A \neq A \times B$$

- The order of operations matters.

Pre-requisite: Subspace

- Understanding the **subspace** definition is necessary before defining a projection matrix.

2D Subspace Example

- In $\mathbb{R}^2$, a **line crossing through the origin** is a subspace.
- This line is infinitely long.

3D Subspace Example

- In $\mathbb{R}^3$, a **plane crossing through the origin** is a subspace.
- This plane spreads to infinity.

Projection Matrix Goal

- Our computer calculates a projection matrix corresponding to a subspace.
- This matrix moves **every vector outside the subspace onto the subspace**.

2D Projection Visualization

- Example: Projecting onto a line (1D subspace).
- Every dot is **compressed onto the line**.

3D Projection Visualization

- Example: Projecting onto a plane (2D subspace).
- This creates a similar vibe of **compression** to a lower dimension.

Defining Projection

- The process is called projection because every vector always moves to its **closest point** on the subspace.

Closest Point Example

- The vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is outside the line.
- The projection matrix moves it exactly to its closest point on that line.

Universal Target

- All vectors outside the subspace are targeted by the projection matrix and moved to their **closest landing spot**.

Analogy: Dimensional Compression

- This type of transformation is similar to **compressing a higher dimension** into a lower one.
- This results in a loss of information.

The Question of Untransformation

- After a transformation, is there one matrix that can move every vector back to where it originally came from?

The Inverse Matrix

- The short answer is no universal matrix exists.
- But, for **every matrix**, an **inverse matrix** can be calculated to perform that **particular untransformation**.

Simple Inverse Examples

- A matrix that scales has an inverse that **un-scales**.
- A matrix that reflects has an inverse that **reflects back**.
- A matrix that rotates has an inverse that **rotates back**.

Complex Inverse Example

- A matrix that shears has an inverse that **un-shears**.
- The inverse matrix always restores all vectors to their original position.

The Identity Result

- Applying a matrix and then its inverse is like **nothing happened**.
- When you compose the two transformations, you get the transformation of **no transformation** (the **Identity Matrix**).

Non-Invertible Matrices

- Some matrices **cannot be inverted** (they are called **singular**).

Examples of Non-Invertible

- The **Zero Matrix** and the **Projection Matrix** cannot be untransformed.
- This is because the process of projection involves a **loss of information** (squashing from higher to lower dimension).

The Nature of a Matrix

- Does a matrix intrinsically carry a visual value? **No.**
- Visualization is an artificial human construct to make sense of the concepts.

Different Interpretations

- The interpretation of a matrix belongs to the user:
 - ▶ **Circuit Students:** A tool to solve system of equations.
 - ▶ **Probability Students:** Representation of a **Markovian process**.
 - ▶ **Data Scientists:** A manifestation of a **data table**.
 - ▶ **Deep Learning:** A Python function that takes a vector input and returns a vector output.

Value of Visual Interpretation

- 1 It provides **intuition** about matrix transformation on vectors.
- 2 It is the direct foundation for **Computer Graphics**.
- 3 It is the necessary gateway to understanding **Matrix Decomposition**.

Looking Ahead: Chapter 2

- Complex transformations can be **re-expressed** into a sequence of **simple transformations** (e.g., rotation, scaling).
- This is the focus of **Matrix Decomposition**.
- Chapter 2 introduces **Spectral Decomposition**, "The King of all Matrices."

SVD: The Grand Finale

- **Singular Value Decomposition (SVD)** combines all important concepts of linear algebra into one theorem.
- It is extremely relevant in **data science** and **machine learning**.

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The Essence of SVD

- **Any Matrix (A):** Regardless of symmetry, rank, or shape, it can be decomposed.
- **Unconditional Decomposition** into three very special matrices.

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Prerequisites for SVD

- An intuitive understanding of:
 - ▶ **Diagonal Matrix** (stretches axes).
 - ▶ **Orthogonal Matrix** (produces rotation).
- The main appeal of SVD is that it generalizes to **rectangular matrices**.
- A vector in R^2 (e.g., $(1, 2)$) is fundamentally different from a vector in R^3 (e.g., $(1, 2, 0)$).
- The R^2 vector is **always devoid** of the third dimension.
- A rectangular matrix can transform a vector from one dimension to another.
- Example: A 2×3 matrix transforms a vector in R^3 **down** to a vector in R^2 .

Visualizing the Simplest Case

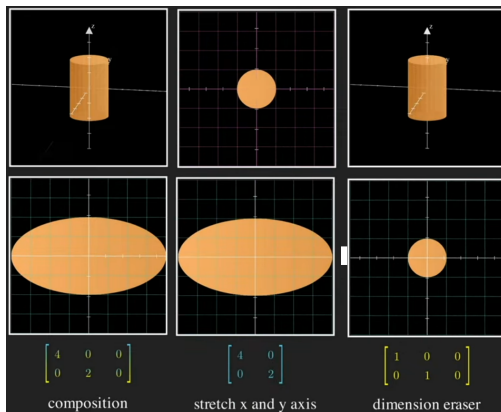
- Consider the "rectangular identity matrix" or **Dimension Eraser**.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

- This matrix transforms $R^3 \rightarrow R^2$.
- It **preserves X and Y** but completely **removes the Z value**.
- Vector $(1, 2, 1)$ in R^3 is mapped down to $(1, 2)$ in R^2 .
- ALL vectors of the form $(1, 2, \mathbf{Z})$ transform to $(1, 2)$.

Composition with Rectangular Matrices

- Multiplying the Dimension Eraser with a Diagonal Matrix:
- $C = D \times \text{Eraser}$
- C first removes the 3rd dimension, then stretches the X and Y axes.
 C encapsulates both.



Symmetric Matrix Flashback

- A **Symmetric Matrix** is a square matrix where entries across the diagonal are identical ($A = A^T$).
- Its eigenvectors are guaranteed to be **perpendicular** (orthogonal).
- Normalizing and packaging the perpendicular eigenvectors creates an **Orthogonal Matrix**.
- An orthogonal matrix implies **rotation**.

Naming the Symmetric Products

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

$A_{2 \times 3} \qquad A^T_{3 \times 2} \qquad AA^T_{2 \times 2}$

- For a 2×3 matrix A :
- $S_{\text{left}} = AA^T$ (a 2×2 symmetric matrix).
- S_{left} has **two perpendicular eigenvectors** (in R^2).
- Eigenvectors of S_{left} are the **Left Singular Vectors** of A .

Naming the Symmetric Products

$$\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}$$

$A^T \quad 3 \times 2 \qquad A \quad 2 \times 3 \qquad A^T A \quad 3 \times 3$

- For a 2×3 matrix A :
- $S_{\text{right}} = A^T A$ (a 3×3 symmetric matrix).
- S_{right} has **three perpendicular eigenvectors** (in R^3).
- Eigenvectors of S_{right} are the **Right Singular Vectors** of A .

Fact 1: Eigenvalues are Non-Negative

- S_{left} and S_{right} are Positive Semi-Definite (PSD) matrices.
- This implies all of their **eigenvalues are non-negative**.
- If the eigenvalues for both S_{left} and S_{right} are sorted in descending order:
 - The overlapping ones are **numerically identical**.

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The SVD Theorem

- Any matrix A can be unconditionally decomposed as:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

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The Matrices' Properties

- A : The original matrix (size $M \times N$).
- Σ : Rectangularly diagonal matrix (size $M \times N$).
- U and V : **Orthogonal Matrices**.

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Matrix Σ (Sigma)

- Σ has the **same dimension** as A .
- Its diagonal entries are the **singular values** of A , arranged in **descending order**.
- All other entries are zero.

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Matrix U

- U contains the **normalized eigenvectors** of $S_{\text{left}} = AA^T$.
- These are the **Left Singular Vectors** of A .
- Columns are arranged in descending order of their corresponding eigenvalues.

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Matrix V^T (V Transpose)

- V contains the **normalized eigenvectors** of $S_{\text{right}} = A^T A$.
- These are the **Right Singular Vectors** of A .
- V is arranged in descending order, and then **transposed** to get V^T .

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SVD as Composition: The Goal

- A single, complicated linear transformation by matrix A can be perfectly understood as the **sequential application** of the three simple matrices:

$$\mathbf{A} \rightarrow \mathbf{V}^T \rightarrow \mathbf{\Sigma} \rightarrow \mathbf{U}$$

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Step 1: Rotation by V^T

- V^T is an **orthogonal matrix** (rotation).
- It rotates the right singular vectors to align with the **standard basis** vectors.
- The vector corresponding to the largest singular value lands on the X-axis, and so on.

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Step 2: Scaling and Projection by Σ

- Σ is a diagonal matrix combined with a **dimension eraser**.
- The dimension eraser **removes the lowest dimension** (e.g., the 3rd dimension).
- The diagonal part **stretches the axes** (X and Y) based on the singular values.

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Step 3: Final Rotation by U

- U is an **orthogonal matrix** (rotation).
- It rotates the standard bases to align with the **left singular vectors**.

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The Spheres-to-Ellipsoids Map

- SVD shows that **any matrix** maps a **sphere** into an **ellipsoid** across different dimensions.

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Visualization Step 1: V^T (Rotation)

- The first step is always a rotation in R^N .
- Rotating a sphere results in a **sphere** (geometry is preserved).

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Visualization Step 2a: Σ (Dimension Change)

- For $R^3 \rightarrow R^2$ (as in the example), the dimension eraser takes away the third dimension.
- A sphere in R^3 becomes a **circle** in R^2 .

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Visualization Step 2b: Σ (Scaling)

- The diagonal part stretches the circle.
- Stretching a circle (non-uniformly) results in an **ellipse**.

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Visualization Step 3: U (Final Rotation)

- The final rotation preserves the geometry of the ellipse.
- It changes only its **position** in R^M .

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Example: $R^2 \rightarrow R^3$ (Addition)

- The spirit of SVD still holds for $R^2 \rightarrow R^3$ transformations.
- Begins with rotation (V^T) in R^2 .

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Example: $R^2 \rightarrow R^3$ (Scaling & Addition)

- Σ scales based on singular values.
- Σ then acts as a **Dimension Adder** (appending a zero dimension).

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Example: $R^2 \rightarrow R^3$ (Final Rotation)

- Final rotation (U) aligns the standard bases (X, Y, Z) with the left singular vectors.

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The True Purpose of SVD

- Is SVD merely the decomposition into four sequential actions (V^T, Σ, U)?
- **Not exactly.** The meaning of a matrix changes under different contexts (visualization is just one context).

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Alternative Interpretation: Rank-One Summation

- A popular interpretation is viewing a high-rank matrix as a **summation of rank-one matrices**.

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots$$

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Application: Low Rank Approximation

- A picture is a massive matrix of pixels.
- SVD enables **low rank approximation** (data compression).
- The picture is gradually approximated by adding more rank-one matrices.

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Insight, Not Numbers

- "The purpose of computing is **Insight, not numbers.**"
- SVD as a pure math exercise can be confusing.

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The Value of Visualization

- Decomposing A into distinct pieces ($U\Sigma V^T$) gives greater insight.
- Every single step is **interpretable** and makes perfect sense.
- We know **precisely** what is going on.

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Connection to PCA

- The initial step of rotation (V^T) of the right singular vectors captures the essence of **Principal Component Analysis (PCA)**.

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Intuition Beyond 3D

- Even in higher dimensions, this decomposition gives reasonable intuition.
- **Rotation** (V^T or U) does not scale or squash objects, only changes their orientation.
- The **Dimension Eraser** (Σ) accounts for the removal of dimensions.

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SVD: A Monumental Theorem

- SVD shares a similarity with the **Fourier Transform** (another monumental theorem).
- Both theorems share an obsession with extreme **generalization** and **decomposition**.
- **Fourier**: Any function $\rightarrow$ sum of sinusoidal waves.
- **SVD**: Any matrix $\rightarrow$ sequence of rotation, scaling, and rotation.

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